science	name	type	people	overview	total	impact	publications	toys	access	reasoning	w	w	w
Computing	Advanced Robotics & Automation (ARA) Lab	Lab	Hung Manh La (Director)	Autonomous robotic systems for civil infrastructure inspection; climbing robots, multi-robot control, UAV flocking; 2025 ASCE Pankow Award	70	690	A climbing robot for steel bridge inspection A practical climbing robot for steel bridge inspection Visual and 3D mapping for steel bridge	Magnetic-wheel climbing robots, UAVs/drones, GPR, seismic/acoustic arrays, laser scanners, IMU, Intel NUC, ROS	Project-based; inquire through lab	Hands-on robotics; award-winning bridge inspection robots; field deployment	www	www	www
Computing	BRAIN Core (Bridge to Research in AI and Neurocomputing)	Core Facility	Alireza Tavakkoli (Director)	UNR core providing computational support for biomedical research: machine learning, AI, statistical analysis, VR/AR platforms for neuroscience investigators.	200	2048	Machine learning applications in neuroscience research AI-driven biomedical data analysis pipelines Computational methods for	HPC infrastructure, cloud AI systems, ML/DL frameworks, statistical tools, VR/AR equipment	Free consultation; fee-based equipment access;		www	www	www
Computing	Computer Vision Laboratory (CVL)	Lab	George Bebis (Director), Alireza Tavakkoli	Pure and applied research in computer vision since 1998: image segmentation, object recognition, 3D reconstruction, face detection, biometrics, tracking	239	2248	Face detection and verification using genetic search Low-resolution face recognition with single sample per person Face detection in the near-IR spectrum	GPU clusters, image capture/ display devices, robotic peripherals, Pronghorn HPC access	Open applications for research assistants year-round;	Major CV lab; open to undergrads; GPU access; directly relevant to	www	www	www
Computing	Electrical Power & Renewable Energy Systems Lab (EPRESL)	Research Lab	Mehdi Etezadi-Amoli, M. Sami Fadali, Yantao Shen	Power systems modeling, renewable energy integration, smart grid, electric vehicle charging infrastructure, and power electronics research.	68	393	Optimal scheduling of electric vehicle charging stations with renewable energy Smart grid integration of distributed generation systems	Power systems simulators, SCADA systems, PV arrays, battery storage testbeds, power electronics test benches, real-time digital simulators	Through Electrical & Biomedical Engineering; graduate		www	www	www
Computing	Evolutionary Computing Systems Lab (ECSL)	Lab	Sushil Louis (Director)	Evolutionary computing, genetic algorithms, AI for RTS games, Case-Injected Genetic Algorithms (CIGAR)	25	99	Playing to learn: case-injected genetic algorithms for computer games Trap avoidance in strategic game playing with case injected genetic algorithms	Computing systems, ML frameworks, game engines, optimization software	Through graduate programs and research assistantships		www	www	www
Computing	HPCVIS Lab	Lab	Frederick Harris Jr. (Co-director), Phillip Goodman	Graphics, scientific visualization, virtual reality applications, and high-performance computation including immersive VR environments	87	344	VFire: immersive wildfire simulation and visualization Open-source tools for immersive environmental visualization A Unity framework for multi-user VR	GPU clusters, VR headsets (Oculus, HTC Vive), Unity VR framework, visualization hardware	Faculty-led; undergrad/grad involvement through CSE department		www	www	www
Computing	Machine Perception Lab	Lab	Emily Hand (Director)	Wearable devices for social interaction support (autism, visual impairment); bridges human and computer vision using ML	28	120	Improving face recognition using artistic interpretations of prominent features Transformation of face recognition technology and representation for faces Empirical study of face recognition	Wearable prototyping, CV systems, ML frameworks, GPU clusters	Open to grad and undergrad students; part of CVL community	ML/vision with social impact; NASA background PI; open to undergrads	www	www	www
Computing	Pronghorn HPC Cluster	Infrastructure	UNR Cyberinfrastructure group	Shared high-performance computing: 108 CPU nodes (3,456 cores), 44 NVIDIA P100 GPUs, 3 V100 GPUs; hosted at Switch data center	0	0		3,456 CPU cores, 24.8 TiB memory, 44 Tesla P100 GPUs, 3 Tesla V100 GPUs, high-speed interconnect	UNR employees: email hpc@unr.edu; external:	Central GPU/HPC resource for ML and computational research across all sciences	www	www	www
Physics	CTEMPS (Center for Transformative Environmental Monitoring Programs)	Research Center	Scott Tyler (Director), Christine Hatch	NSF-funded center pioneering distributed temperature sensing (DTS) fiber optic technology for environmental monitoring of rivers, glaciers, caves, and subsurface systems.	31	203	Environmental temperature sensing using Raman spectra DTS fiber-optic methods Distributed temperature sensing for hydrology and environmental	Fiber-optic DTS instruments, environmental sensors, weather stations, data loggers, field deployment systems	Equipment lending program for researchers; NSF-supported;	Pioneered DTS technology for environmental science; global impact	www	www	www
Physics	Center for Civil Engineering Earthquake Research (CCEER)	Research Center	Sherif Elfass (Director), Ian Buckle, Saiid Saiidi, Patrick Laplace	Nation's premier university earthquake engineering lab with 5 shake tables including the largest Laminar Soil Box (350 tons). Research in structural, geotechnical, and bridge earthquake	16	113	Seismic performance of reinforced concrete bridge columns under combined loading Laminar soil box testing of soil-foundation-structure interaction	5 shake tables, Laminar Soil Box (350 tons), strong floor, reaction wall, 300-kip actuators, data acquisition systems, SAGE building (153,000 sq ft)	Graduate research through Civil & Environmental Engineering;	Unique facility in US university system; 5 shake tables; world-class earthquake	www	www	www
Physics	Derevianko Group	Research Group	Andrei Derevianko (AAAS Fellow, Simons Fellow)	Novel atomic clock development; precision tests of fundamental symmetries; ytterbium optical lattice clocks; nuclear clocks; parity-non-conservation	64	614	Hunting for topological dark matter with atomic clocks Physics of optical lattice clocks Search for new physics with atoms and molecules	Precision laser systems, optical lattice clock apparatus, high-precision spectroscopy	Grad/undergrad through Physics dept	World-leading precision measurement; AAAS Fellow; atomic clocks are	www	www	www
Physics	Great Basin Center for Geothermal Energy	Research Center	Bridget Ayling (Director), James Faulds, Nicholas Hinz	DOE-funded geothermal research center advancing exploration, characterization, and development of geothermal resources in the Great Basin and globally.	99	1152	Controls on Great Basin geothermal systems Geothermal play fairway analysis of the Great Basin Machine learning approaches for	Geothermal well logging equipment, geophysical instruments, thermal modeling software, GIS systems, field vehicles	Through Geological Sciences & Engineering; DOE-funded	DOE-funded; unique geothermal expertise leveraging Nevada's geologic	www	www	www
Physics	High Energy Density Lab (HEDL)	Research Group	Thomas White	Laboratory astrophysics; extreme states of matter; equation of state in planetary cores; uses NIF, LCLS, European XFEL	58	1441	Accessing extreme equation of state on National Ignition Facility Probing matter at Gbar pressures at the NIF Developing absolute shock wave	Access to National Ignition Facility, LCLS X-ray laser, European XFEL, DFT/MD computing	Through White research group; DOE funded	Access to world-leading national facilities (NIF, LCLS); bridges physics +	www	www	www
Physics	Nevada Geodetic Laboratory	Research Lab	Geoffrey Blewitt (Director), Corné Kreemer, William Hammond	World-leading GPS/geodesy research operating 414-station GPS network. Real-time crustal deformation, tectonic strain analysis, and sea-level change monitoring since 2004.	75	1240	Harnessing the GPS data explosion for interdisciplinary science A global strain rate model including plate boundary zones GPS measurement of present-day	414-station GPS network, GNSS processing systems, InSAR data, high-performance computing, geodetic data archives	Public data at geodesy.unr.edu; research through Nevada Bureau of	World-leading GPS geodesy lab; highest-impact geoscience group at UNR; open	www	www	www
Physics	Tscherbul Group	Research Group	Timur Tscherbul (NSF CAREER)	Ultracold quantum dynamics: molecular collisions, ultracold chemistry, photosynthetic light-harvesting, Rydberg aggregates, superfluid helium nanodroplets	72	313	Hyperfine-to-rotational energy transfer in ultracold collisions A rigorous adiabatic approach to ultracold collisions A Feshbach resonance in triplet ground-	Quantum computing resources, HPC (coupled-channel methods, quantum master equations, Monte Carlo)	Grad/undergrad through Physics dept		www	www	www
Physics	Weinstein Lab	Lab	Jonathan Weinstein	Cryogenic cooling and laser spectroscopy of atoms/molecules trapped in solid neon and parahydrogen at near-absolute-zero; quantum sensing; dark matter searches	5	37	Spin coherence and optical properties in solid parahydrogen Enhanced spin coherence of rubidium atoms in solid neon Ultralong spin coherence times via	Cryostats, laser systems, optical tables, vacuum chambers, magnetic shielding, spectrometers, matrix isolation apparatus	Grad/undergrad through Physics dept	Quantum sensing and fundamental physics; Harvard-trained PI; bridges physics and	www	www	www
Physics	Zebra Pulsed Power Lab (ZPPL)	Lab	Thomas White (lead), Roberto Mancini, Hiroshi Sawada, Bernhard Bach	Highest-power pulsed electrical device at any US university; Z-pinch physics, HED science, laser-plasma interactions; 153,000 sq ft SAGE facility	96	1694	Two-terawatt Zebra Z-pinch at Nevada terawatt facility Staged Z-pinch experiments at 1MA Zebra generator Measurement of ionization state of wire-	Zebra (2 TW pulsed power), Leopard laser (50 TW, 300 fs), Falcon optical compressor, X-ray diagnostics, streak cameras, ICCD cameras, 800-	Grad/undergrad through Physics dept; DOE/ NNSA supported;	World-class pulsed power facility; unique in US university system; Core diagnostics.	www	www	www
Astronomy	Great Basin Observatory (GBO)	Observatory	Richard Plotkin (UNR PI), multi-institutional	0.7-meter robotic remotely-operated telescope in Great Basin National Park (elev. 6,825 ft, 70+ mi from light pollution); NSF-funded astrophysics and K-12 education	69	335	User-guided photometric pipeline for GBO robotic telescope Photometry development for GBO telescope Robotic telescope black hole accretion	0.7m research-grade robotic telescope, CCD cameras, photometry pipeline, remote operation software	UNR faculty/ student access as partner institution; K-12 outreach	Research-grade remote telescope; direct astronomy olympiad alignment;	www	www	www
Chemistry	Borotto Research Group	Lab	Nicholas Borotto	Mass spectrometry-based detection of biomolecules; protein structure characterization; post-translational modification analysis	36	168	The path forward for protein footprinting covalent labeling and mass spec Investigating therapeutic protein structure with diethylpyrocarbonate labeling	timsTOF Pro2, ion mobility MS, tandem MS, IRMPD, CO laser-equipped MS, LC separations	Open to grad and undergrad students	Advanced mass spec directly relevant to SIL instruments and proteomics.	www	www	www
Chemistry	Fluid Inclusions Lab	Research Lab	Matthieu Harlaux (Director, Economic Geology)	Microthermometric and microanalytical study of fluid inclusions in minerals for understanding hydrothermal ore deposit genesis, geothermal systems, and geological processes.	24	209	Geochemical signature of magmatic-hydrothermal fluids in rare-metal granites Sequential LA-ICP-MS for fluid inclusion trace element analysis	Linkam heating/freezing stages (-200 to +600°C and up to 1400°C), Raman spectrometer, Olympus BX51 microscope with NIR video, LA-ICP-MS	Through Geological Sciences; courses GEOL 713/776;	Toys list.	www	www	www
Chemistry	Geary Lab	Lab	Laina Geary	Palladium-catalyzed cross-coupling, C-H functionalization, Pauson-Khand cycloadditions, alcohol substitution for C-C/C-X bonds	5	3	Recent advances in the Pauson-Khand reaction Nitrous oxide-promoted Pauson-Khand cycloadditions Regioselectivity in Pauson-Khand	GC-MS, LC-MS, NMR, column chromatography, Schlenk lines, glovebox	Open to grad and undergrad students	Hands-on organic synthesis; uses SIL GC-MS/LC-MS for characterization	www	www	www
Chemistry	Hitchcock Center for Chemical Ecology	Center	Christopher Jeffrey (Director), Lora Richards, Matthew Tucker	20+ faculty across chemistry, biology, ecology, agriculture; studies plant defensive chemistry, insect diversity, bio-prospecting in Great Basin; $3M NSF NRT grant	59	328	Phytochemical diversity drives plant-insect community diversity Structure revision of scytonemin imine and its relationship to scytonemin Isolation of new neolignans and	Shared analytical instruments (LC-MS, GC-MS, NMR), synthesis labs, field collection equipment, metabolomics pipelines	NRT fellowships (25+ supported); open to grad	Premier interdisciplinary center bridging chemistry + biology; NRT	www	www	www

Chemistry	Materials Characterization Nevada (MCNV)	Shared Facility	Dev Chidambaram, Krista Carlson (contacts)	Advanced materials characterization facility housing world's second lab-based EXAFS system. XRD, XRF, thermal analysis, surface characterization for materials science and engineering	95	456	EXAFS characterization of novel battery electrode materials XRD phase analysis of advanced ceramic composites Thermal stability of nanostructured	Lab-based EXAFS system, XRD, XRF, DSC/TGA, surface area analyzers, nanoindenter, profilometer	Fee-for-service through NCAR; open to UNR, external, and industry users	World's second lab EXAFS; complements SIL chemistry capabilities. Toys	www	www	www
Chemistry	Microbeam & SEM Laboratory (Mackay School)	Core Facility	Joel DesOrmeau (Manager), Krista Carlson	State-of-the-art microanalysis facility with NSF-funded field-emission EPMA (first in US), FE-SEM, and LA-ICP-MS for geological, materials, and environmental sample analysis at micrometer resolution.	65	356	GHR1 Zircon: a new natural reference material for microbeam U-Pb geochronology Rapid exhumation of Papua New Guinea UHP terrane via eclogite	JEOL JXA iHP200F FE-EPMA (first in US), JEOL JSM 7100FT FE-SEM, LA-ICP-MS, 5 WDS spectrometers, hyperspectral CL, EDS	Fee-for-service; open to UNR, other universities, and industry;	First FE-EPMA installed in US; NSF-funded; critical analytical infrastructure. Toys	www	www	www
Chemistry	Mineral Processing Lab	Research Lab	Pengbo Chu, Barbara Wold (Mining & Metallurgical Engineering)	Ore processing, flotation chemistry, hydrometallurgy, and mineral characterization research supporting Nevada's mining industry. Part of Mackay School.	32	445	Flotation optimization for complex sulfide ores Hydrometallurgical recovery of rare earth elements Mineral liberation analysis using	Flotation cells, ball mills, jaw crushers, leaching reactors, mineral liberation analyzer, assay furnaces	Through Mining & Metallurgical Engineering; industry partnerships	Toys list.	www	www	www
Chemistry	Shared Instruments Laboratory	Core Facility	SIL / Chemistry Department	Liquid-state and solid-state NMR for structural characterization of organic, inorganic, and materials samples	0	0		3x Agilent Varian liquid-state NMR (including 400-MR with auto-switchable probe: 1H, 19F, 13C, 31P), Tecmag Discovery 400 MHz solid-state NMR with	Through SIL; training required; iLab scheduling	NMR is essential for organic/ inorganic chemistry characterization;	www	www	www
Chemistry	Ralph Roberts Center for Economic Geology (CREG)	Research Center	John Muntean (Director, Arthur Brant Endowed Chair)	30-year partnership between Mackay School, Nevada Bureau of Mines & Geology, mining industry, and USGS. Focuses on Carlin-type gold deposit genesis and mineral exploration methods.	4	2	Diversity of Carlin-Style Gold Deposits: applications to exploration The Carlin gold system: exploration applications in Nevada Geochemistry of metasedimentary rock-	Analytical lab access (SEM, EPMA, LA-ICP-MS at Mackay), field exploration equipment, mining industry partnerships, USGS collaboration	Graduate student thesis funding; mining industry internships;	Premier economic geology center; 30-year mining industry partnership. Toys	www	www	www
Chemistry	Shared Instruments Laboratory	Core Facility	Stephen Spain (Director)	Free-access core facility with mass spec, vibrational spectroscopy, thermal analysis, optical spectroscopy, chromatography; Room 016 Chemistry Building				LC-MS (Waters + Agilent TOF), MALDI-8020, Raman microscope, 2x FTIR (Nicolet 380 + Bruker Tensor 27), GC-MS, DSC, TGA, fluorescence,	Free for UNR students/faculty after training; 24/7 card access;	You already have 24/7 access; comprehensive instrument suite across	www	www	www
Chemistry	Tal-Gan Research Group	Lab	Yftah Tal-Gan	Peptide-based tools for bacterial quorum sensing; therapeutic implications for bacterial pathogenesis and virulence	35	1900	Highly potent inhibitors of quorum sensing in Staphylococcus aureus Targeting peptide-based quorum sensing systems for Gram-positive treatment	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture, molecular microbiology setup	Open to undergrad, grad, and postdocs; NIH/ NSF funded	Chemical biology; peptide synthesis; uses SIL LC-MS and MALDI; open to undergrads	www	www	www
Chemistry	Tucker Research Group	Lab	Matthew Tucker	Ultrafast 2D IR spectroscopy of protein folding, binding, and function; also studies magnetism and energy dissipation in desert lichen molecules	6	80	Nonequilibrium dynamics of helix reorganization observed by transient 2D IR Site-specific coupling of hydration water and protein flexibility via 2D-IR	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	Open to grad and undergrad students	FTIR/spectroscopy directly relevant to SIL instruments; biophysics bridge. Toys list.	www	www	www
Chemistry	X-ray Crystallography Facility	Core Facility	SIL / Chemistry Department	Single-crystal X-ray diffraction for molecular structure determination	0	0		Bruker Smart APEX diffractometer	Through SIL; sample submission; specialist operation		www	www	www
Chemistry	Yang Research Group	Lab	Ying Yang (NSF CAREER)	Dynamic polymers with reversible bonds for recyclable plastics, self-healing materials, 3D printing, wearable sensors, soft robotics	87	2082	A stretchable mechanically robust polymer with shape-memory-assisted healing Intrinsic self-healing polymers with high E-modulus based on dynamic urea	Polymer synthesis equipment, 3D printers, mechanical testing, spectroscopic characterization	Open to grad and undergrad; 2025 Foundation Early Career	New faculty building lab; dynamic materials are hands-on; likely open to	www	www	www
Chemistry	de Bettencourt-Dias Group	Lab	Ana de Bettencourt-Dias	Luminescent lanthanide complexes; f-element chemistry for energy efficiency, diagnostics, therapy	28	2400	Luminescent lanthanide complexes with dual activity for photodynamic therapy Lanthanide luminescence for biomedical analyses and imaging Ligand design in lanthanide complexes	Fluorescence spectrometers, UV-Vis, X-ray crystallography, glovebox, NMR, CD spectrometer	Open to grad and undergrad students		www		www
Biology	Aquatic Ecosystems Analysis Lab	Lab	Sudeep Chandra, Zeb Hogan	Limnology, aquatic conservation, invasive species management, freshwater fish ecology; Castle Lake station (est. 1959)	114	2643	Lake Tahoe invasive species management and control Aquatic ecosystems conservation and food web ecology Invasive species impacts on aquatic	Water quality probes, CTD profilers, boats, field sampling, isotope analysis	Open to undergrad/grad; Castle Lake has trained 700+		www	www	www
Biology	Cellular and Molecular Imaging Core	Core Facility	Alexander van der Linden (Director), Andrew Yanez (Training	Part of Center for Integrative Neuroscience COBRE. Advanced microscopy, DNA/RNA/protein analysis, and cellular imaging on Fleischmann 3rd floor.	8	40	Confocal microscopy applications in neuroscience Two-photon imaging of neural tissue architecture Tissue clearing and 3D imaging	Leica SP8 confocal with LIGHTNING, Scientifica two-photon, Leica THUNDER 3D, Typhoon Phosphoimager, PCR robot, real-time PCR, cell	UNR researchers; iLab reservation system; training available	Toys list.	www	www	www
Biology	Center for Integrative Neuroscience (COBRE)	Research Center	70+ affiliated faculty across 13 departments	NIH COBRE-funded center (P30 Phase 3, $5.3M) since 2012. Operates molecular imaging, neuroimaging, and VR/AI cores. Supports interdisciplinary PhD in Integrative Neuroscience.	220	8800	Multi-modal neuroimaging studies of brain network organization Neural circuit mechanisms of sensory processing Computational neuroscience	Confocal microscope (Leica SP8), two-photon microscope, THUNDER 3D imaging, VR/AR systems, fMRI, EEG, $9M invested in equipment	70+ faculty in 13 departments; pilot project funding; PhD	Major NIH-funded neuroscience center; $9M equipment investment; Phase	www	www	www
Biology	Center for Molecular Medicine (UNR Med)	Research Center	Multiple department heads (Microbiology, Pharmacology,	107,000 sq ft research facility housing wet labs, rodent vivarium, BSL3 select agent lab. Programs in cancer biology, infectious disease, cardiovascular disease, and neurosciences.	120	4800	Cell biology signaling mechanisms in cardiovascular disease Neuro-immune disease pathogenesis and biomarkers Infectious disease mechanisms in BSL3	BSL3 select agent lab, rodent vivarium, wet labs, medical office suites, cell culture facilities, microscopy	UNR Med faculty and students; partnership with Renown Health	Major UNR Med research hub; BSL3 capability; 107,000 sq ft	www	www	www
Biology	Global Water Center	Center	Sudeep Chandra (Director)	Multidisciplinary center for water sustainability; research on every continent; translates science into actionable solutions	72	1802	Lake Tahoe invasive species management and control Water sustainability and aquatic ecosystems conservation Global freshwater solutions for	Water quality instruments, drone technology, monitoring equipment, predictive modeling	Open to faculty/ students across disciplines; NGO partnerships		www	www	www
Biology	Great Basin Bug Lab	Lab	Matt Forister	Plant-insect ecology, butterfly ecology, diet breadth evolution, phenology, conservation; Science 2025 paper on US butterfly declines	7	54	Increasing neonicotinoid use and butterfly fauna decline Dramatic decline in western butterfly populations linked to pesticides Climate change is a stressor of insect	Field collection equipment, specimen collections, molecular tools, genomic sequencing, ecological software	Open to PhD, Master's, and undergrad students; field + lab research	Very high impact PI (Science 2025); accessible fieldwork; biodiversity +	www	www	www
Biology	Institute for Neuroscience	Institute	Jennifer Hoy (Director), Dennis Mathew, multiple faculty; 70+ affiliated	Multidisciplinary neuroscience center; $25M NIH COBRE funding across 3 phases; 40 active grad students; doctoral program	21	93	Defined cell types in superior colliculus prey capture Layer-specific refinement of visual cortex after eye opening Identification of a brainstem circuit	HD EEG, 2x 3T MRI (at Renown), NIRS, Leica SP8 confocal, Scientifica 2-photon microscope, Leica THUNDER 3D, VR/AR equipment	Grad students in Integrative Neuroscience Program; core facilities	Major neuroscience institute; 2-photon and confocal microscopy;	www	www	www
Biology	Mathew Lab	Lab	Dennis Mathew	Neuroscience of olfaction in Drosophila; neural circuit function; neuromodulation; NIH ENDURE training program	12	52	Internal state affects local neuron function in olfactory behavior Behavior responses to chemical and optogenetic Drosophila stimuli Differential contributions of olfactory	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	Open to grad and undergrad; runs NIH training program; co-	Drosophila accessible model; NIH training program for undergrads; neuro	www	www	www
Biology	Nevada INBRE (IDeA Network of Biomedical Research Excellence)	Research Network	NIH-funded program directors, multi-institution faculty	NIH program building biomedical research infrastructure across Nevada. Supports early-stage investigators, research labs, data science core, and undergraduate pipeline programs at 5+ institutions.	180	3600	Bioinformatics and computational biology methodologies for Nevada researchers Biomedical research capacity building in EPSCoR states	Bioinformatics servers, computational platforms, data science infrastructure, research funding	Faculty and students at participating Nevada institutions;	Statewide biomedical research infrastructure; NIH-funded. Core	www	www	www
Biology	Ouyang Lab	Lab	Jenny Ouyang	Urban ecology, artificial light pollution effects on birds, circadian biology, physiological adaptation; NSF CAREER $1.2M	37	244	Restless roosts: light pollution affects behavior and sleep physiology Effects of dim artificial light on circadian clock genes in songbird Birds of a feather flock together: social	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	Open to grad and undergrad students	NSF CAREER; accessible field biology; bridges biology + physics via light/circadian	www	www	www
Biology	Parchman Lab	Lab	Thomas Parchman	Population genomics: adaptive divergence, hybridization, speciation in conifers, tropical Piper, birds, and fish	57	429	Genome divergence and diversification in geographic mosaic of coevolution Coevolution, local adaptation, and ecological speciation in birds Resource stability and geographic	Molecular biology lab, genomic sequencing, computational resources for genome-scale data	Open to grad and undergrad students		www	www	www
Biology	Voyles Lab	Lab	Jamie Voyles (McMinn Research Professor)	Emerging infectious diseases in amphibians; chytridiomycosis; conservation biology; field research in Panama since 2003	31	260	Pathogenesis of chytridiomycosis causing catastrophic amphibian declines Amphibian fungal panzootic causes catastrophic ongoing biodiversity loss	Pathogen research facilities, physiological analysis, field stations (Panama: Smithsonian SCBI)	Open to grad and undergrad; international field opportunities		www	www	www

Core Facility	Cell Analysis Core (Flow Cytometry)	Core Facility	School of Medicine	Flow cytometry and cell sorting; spatial molecular imaging; cell cycle analysis, proliferation assays	0	0		WOLF microfluidics sorter, Cytoflex 3-laser cytometer, CosMx spatial molecular imager, FlowJo	Fee-based; iLab scheduling; School of Medicine	Toys list.	www		www
Core Facility	Core Analytical Laboratory (CAL)	Core Facility	College of Agriculture + USDA ARS	Analytical chemistry for agricultural, environmental, and plant science; elemental analysis, chromatography	0	0		ICP-AES, ICP-MS, GC (MS/FID/ECD), elemental analyzer, flow injection, UV-Vis, microwave digestion, particle size	Students can train to operate independently; staff can analyze for	Toys list.	www		www
Core Facility	Davidson Cleanroom / Nanofab Lab	Core Facility	Russ Renzas (Director)	Class 1,000 semiconductor/photonic fabrication; lithography, deposition, etch, metrology, wire bonding; 2,937 sq ft				Lithography systems, deposition tools, etching equipment, metrology instruments, wire bonder	UNR researchers; external: industrial rates; Pennington	Nanofabrication for quantum/photonic devices; Physics olympiad alignment. Toys	www		www
Core Facility	Electron Microscopy Facility (EMMF)	Core Facility	Yufeng Zheng (Director)	Scanning and transmission electron microscopy for metals, ceramics, polymers, composites, biological samples	4	1352		Talos F200S S/TEM with EDS, Scios 2 FIB/SEM, FEI Nano600 NanoSEM, sample prep instruments	Contact EMMF; fee-based; iLab scheduling	Electron microscopy gives nanoscale imaging across all sciences;	www		www
Core Facility	Neuroimaging Core	Core Facility	Center for Integrative Neuroscience	MRI/fMRI and neurophysiology; human and small animal imaging	0	0		2x 3T MRI (Philips Ingenia + Siemens Skyra at Renown), 2x HD EEG, NIRS (TechEn)	Available to affiliated faculty/ students; Renown South	Toys list.	www		www
Core Facility	Nevada Bioinformatics Center	Core Facility	Hans Vasquez-Gross (Director)	Bioinformatics services: NGS analysis, single-cell RNA-seq, transcriptomics, genome resequencing, proteomics data, workshops	0	0	Genomic surveillance of Nevada SARS-CoV-2 variants Plant genomics and water-use efficiency in crassulacean metabolism Cellular paraspeckle component in viral	HPC clusters, bioinformatics software, NGS analysis pipelines, computational workstations	Pay-per-service; workshops open to all; iLab scheduling	Toys list.	www		www
Core Facility	Nevada Genomics Center	Core Facility	Andrew Gorzalski (Director)	DNA sequencing core: Illumina NGS, 10X single-cell, Sanger, fragment analysis; serves 132 Nevada labs + 250 labs nationally/internationally	26	341		Illumina NextSeq 2000, MiniSeq, Sanger sequencing, fragment analysis, BSL-2 lab	Pay-per-service; open to all researchers (academic + industry); iLab	Genomics is central to Biology olympiad; accessible pay-per-service model.	www		www
Core Facility	Nevada Proteomics Center	Core Facility	David Quilici (Director)	Mass spectral proteomic analysis; protein identification, quantitation, and PTM analysis from blood, saliva, tissue	8	47	Quantitative proteomics unveils alterations in neuropathic nerves Mass spectrometry analysis for clinical diagnostics Proteomics of physiological pathways	Bruker timsTOF Pro2, ThermoFisher Eclipse MS, Evosep One LC, gel/IP/solution workflows	Pay-per-service; open to all researchers (academic + industry); iLab	Proteomics bridges chemistry + biology; advanced mass spec complements	www	www	www
Core Facility	Nevada Stable Isotope Lab (NSIL)	Core Facility	Simon Poulson	C, O, H, N, S stable isotope analysis for geological, environmental, and biological samples	25	144	Environmental low temperature geochemistry via stable isotopes Stable isotope analysis of carbon, oxygen, hydrogen, nitrogen, sulfur Isotope geochemistry research	Stable isotope ratio mass spectrometers, sample preparation systems	Fee-based through NCAR; ncar@unr.edu	Toys list.	www		www
External	Biosciences Entrepreneurial Lab (BEL)	Incubator	NCAR	Turnkey wet lab incubator (2,200 sq ft) for biotech startups; fully furnished with standard lab equipment	0	0		Complete wet lab infrastructure, standard lab equipment, shared analytical instruments	Through NCAR incubation program; monthly membership or	Toys list.	www		www
External	DeLaMare Makerspace	Maker Space	DeLaMare Science & Engineering Library	3D printing, laser cutting, PCB milling, soldering, prototyping; free for campus community	0	0		3D printers/scanners, laser cutter, vinyl cutter, PCB mill, soldering stations, sewing stations, hand tools	Free for all UNR community; training required; 180	Free access; 3D printing and prototyping for any science project; very accessible.	www		www
External	Desert Research Institute (DRI)	Research Institute	500+ research faculty/staff; Reno + Las Vegas campuses	Nonprofit sister institution to UNR (NSHE); 40+ specialized labs; atmospheric, earth/ecosystem, hydrologic sciences; >$50M annual research	800	35000	Mineral dust aerosol size distribution and Earth system impacts Dust emissions characterization and atmospheric visibility impact Wildfire smoke air quality and aerosol	Atmospheric science instruments, hydrologic measurement, environmental monitoring, LASSO optical/ aerosol equipment	Research collaboration with UNR; some facilities accessible	Major research institution adjacent to UNR; atmospheric science labs	www		www
External	Innevation Center	Innovation Hub	UNR-affiliated	Startup incubator with co-working, makerspace, conference rooms, mentorship; access to NCAR facilities	0	0		Makerspace tools, co-working spaces, conference rooms	Free for UNR students/ faculty/staff; community: paid	Toys list.	www		www
External	NCAR (National Center for Atmospheric Research)	National Research Center	NSF-sponsored (headquarters: Boulder, CO)	Climate science, meteorology, atmospheric chemistry, space weather; no permanent Reno facility but collaborative partnerships with UNR on drought/climate	0	0	NCEP/NCAR 40-year reanalysis project for global climate record Indicators of global climate change and temperature emissions AMOC Atlantic meridional overturning	NCAR-Wyoming Supercomputing Center, Mauna Loa Observatory, field campaign instruments	Collaborative partnerships; no direct Reno presence; data and models		www		www
External	Nevada Center for Applied Research (NCAR)	Applied Research Center	UNR (600+ companies engaged since 2013)	Coordinates access to all 40+ UNR shared research facilities; supports startups; $201M venture capital raised by affiliates	0	0		Access to all UNR core facilities, Biosciences Entrepreneurial Lab (wet lab incubator), Mechanical Engineering Manufacturing Lab	External companies/ researchers: coordinate through	Gateway to all UNR equipment for paid access; NCAR is the main contact for	www		www
External	Nevada Seismological Laboratory	Research Lab	Christie Rowe (Director)	186-station seismic network across Nevada/E. Sierra; 105 fire watch cameras; Earthquake Early Warning; 50+ years of data			Rapid seismic monitoring deployment for earthquake sequence Nevada earthquake detection and monitoring network Earthquake early warning system for	Seismometers (186-station network), fire watch cameras (105), EEW systems, seismic data archives	Research collaboration; data publicly available; Mackay School		www	www	www